Diabetes Risk Prediction by Logistic Regression

Machine Learning Final Project Report

Bishnu Kumar Bishwokarma

Webster University

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Prof. Dipak Biscuitwala

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Executive Summary

The project put machine learning models to the test for predicting diabetes risk, using the Pima Indians Diabetes Database. With 768 records in the database, eight numeric predictors and a binary Outcome variable, we opted for Logistic Regression as our model of choice; it is more readily understood and well suited to a binary outcome. For the sake of comparison other models were employed as well: Linear Discriminant Analysis (LDA), k-Nearest Neighbors (k-NN), Naive Bayes, Random Forest and Gradient Boosted Trees. The whole workflow was done in Altair AI Studio (RapidMiner).

In preparing the data we dealt with zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI by regarding them as missing and substituting the average. We then divided the data into training and testing sets on a 70/30 basis which left 230 records as a holdout set. Since distance-based models require all features to be on the same scale, normalization was put in place for the k-NN portion.

Performance was measured by accuracy, area under the ROC curve (AUC), precision, recall/sensitivity, F1 score, specificity and confusion matrices. On the holdout data the baseline Logistic Regression model provided an overall balance with 76.52% accuracy, an AUC of 0.827, 62.75% precision, 80.00% recall, 70.33% F1 and 74.67% specificity. Random Forest showed a high specificity at 90.67% but its recall was low at 36.25%, missing out on many positive cases. We also ran a grid search with 10fold cross-validation on the Logistic Regression, tuning it with lambda values from 0.001 to 1 on a log scale. While the tuned version remained competitive it only managed 74.35% accuracy and

66.25% recall on the holdout set. All things considered the baseline Logistic Regression was retained as the model. This is an exercise in analysis and not to be taken as a medical diagnosis.

1.Introduction

Diabetes mellitus is a disease that involves blood glucose level and requires correct testing and medication. According to recent guidelines, diabetes can be determined with the help of either A1C level or blood glucose level, and additional testing should be carried out in case of uncertain test results. It was stated by the American Diabetes Association Professional Practice Committee for Diabetes in 2026. However, machine learning cannot be used instead of these guidelines. Rather, it detects patterns in health-related information and predicts the probability of a person belonging to a certain high-risk group. It means that prediction of diabetes is an example of application of data analysis in medicine.

The selected project topic is Diabetes Risk Prediction Using Logistic Regression. The primary objectives include preparing the data on diabetes for modeling developing a Logistic Regression model comparing this model with others such as linear models, probabilistic models, distance models, bagging models and boosting models selecting the optimal model based on performance measures rather than the accuracy alone. The reason for selecting Logistic Regression is due to its efficiency and effectiveness in modeling outcomes and giving statistical results. Pima Indians data set is an example data

set for diabetes studies. In the original study, eight health conditions were considered to predict the diabetes risk among the Pima women.

In our experiment, we created the dataset for classification in two classes. Outcome = 1 indicates that the patient suffers from diabetes while Outcome = 0 shows that he or she does not. The performance of the classifier was assessed using the test set. We report accuracy as well as recall, specificity, F1 score and AUC because in screening scenarios cost of FN and FP errors is not the same.

2. Data Description

2.1 Dataset Overview

The dataset used in this project is the distributed Pima Indians Diabetes dataset. This dataset contains 768 records and nine columns: eight predictor columns and a binary Outcome label. In this dataset there are 500 observations and 268 positive observations so 34.9 percent of the records belong to the positive class. The class distribution in this dataset is moderately imbalanced which is one reason that the evaluation uses more than accuracy.

Variable	Role	Description
Pregnancies	Predictor	Number of pregnancies

Glucose	Predictor	Plasma glucose concentration
BloodPressure	Predictor	Diastolic blood pressure
SkinThickness	Predictor	Triceps skinfold thickness
Insulin	Predictor	Two-hour serum insulin
BMI	Predictor	Body mass index
DiabetesPedigreeFunction	Predictor	Diabetes pedigree function
Age	Predictor	Age in years
Outcome	Target	Binary diabetes class: 0/1 (false/true)

Table 1. Dataset variable and roles.

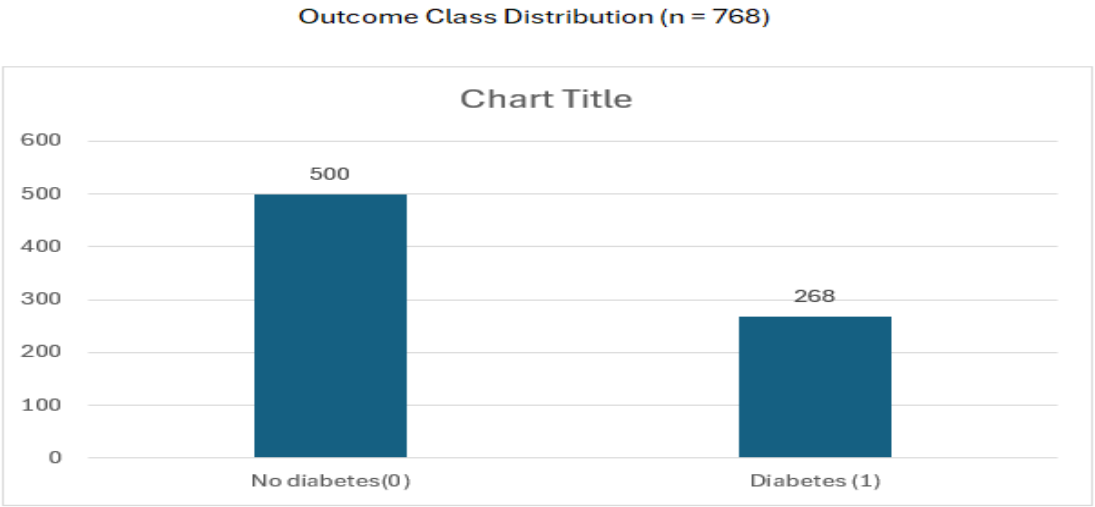


Figure 1. Outcome class distribution in the diabetes dataset.

2.2 Data Preprocessing

The raw dataset had no missing values in the main measure columns, but some variables had zero values which were impossible to occur in the biological world. The procedure thus considered zero values to be missing values for Glucose, BloodPressure, SkinThickness, Insulin, and BMI. The count of these values was 5, 35, 227, 374, and 11 respectively. The missing values were imputed by means of mean, thereby leaving none in the modeling operators. Zero values in the case of Pregnancies were however kept since zero pregnancies are possible, and no change was made in Outcome variable.

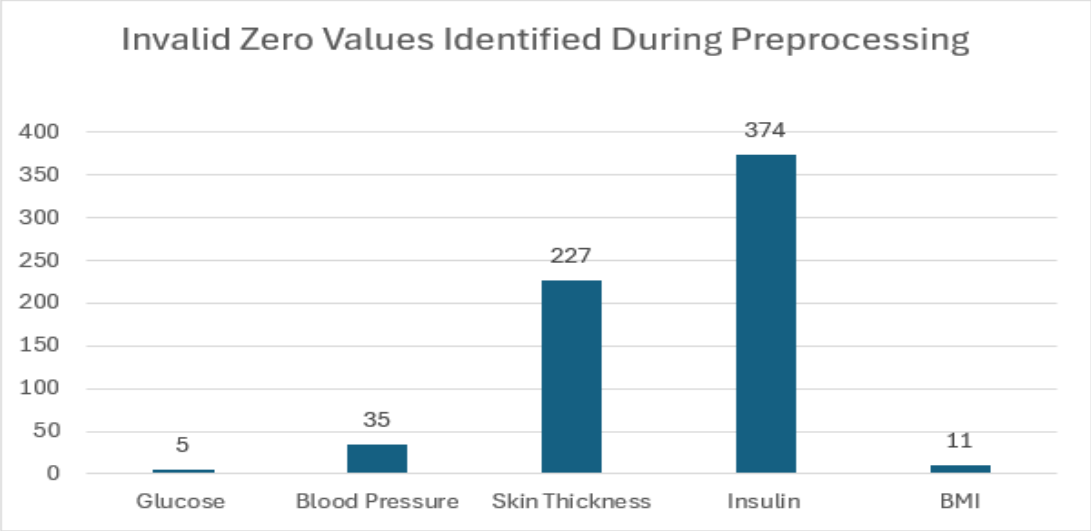


Figure 2. Invalid zero values identified and treated as missing during the preprocessing.

The Outcome field was used as the label and encoded as a bionominal target (false/true). I then divided my cleaned dataset into two parts: 70 percent for training and 30 percent for testing. In total, it gave me 538 training samples and 230 testing samples. I made sure that the ratio of class proportions is preserved in both groups by applying stratified splitting and using a fixed random seed to enable reproducibility of results. Since k-NN uses the measurement of distances between observations, I included normalization as an additional preprocessing step for k-NN to make sure that numeric ranges of variables will not prevail in the process.

3. Exploratory Data Analysis (EDA)

The initial exploratory analysis included the examination of the quality of the data, the balance of classes, and the distribution of health measures. In order to avoid unnecessary cluttering I decided to stress only six features: Glucose, BloodPressure, BMI, Pregnancies, DiabetesPedigreeFunction and Age. SkinThickness and Insulin were available for further modeling after the process of imputation but I decided not to emphasize them due to numerous invalid zeros in the figures.

3.1 Univariate Analysis

Glucose has a range of values in its histogram. The majority of the data points are located near the middle of the range while there are few data points for glucose values. This is because the positive diabetes class has a greater median value of glucose than the negative diabetes class. The boxplot clearly indicates this difference between the two

classes. The histogram is useful for the visualization of the data while the boxplot identifies the difference in the distribution of glucose in the two classes.

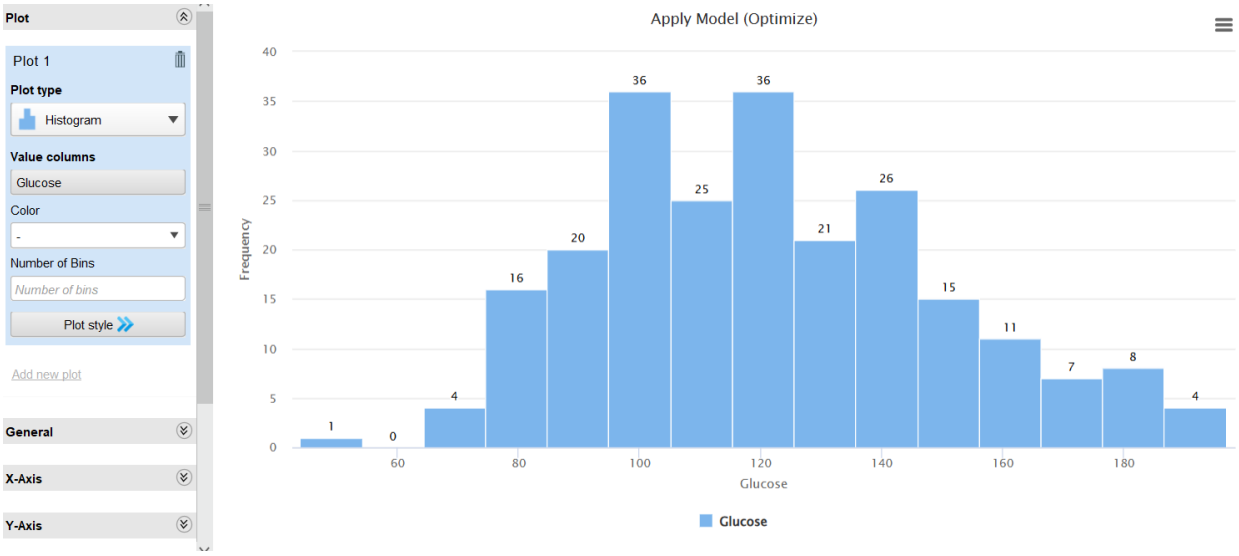


Figure 3. Glucose distribution in the held-out ExampleSet used for reporting.

3.2 Bivariate Analysis

Specific outcomes boxplot comparison shows differences between two of the most explainable predictors. For ExampleSet that is held-out, the median BMI value when Outcome = is equal to 34.75, whereas it equals 30.90 when Outcome = false. Glucose predictor has a higher split, when its median equals 136 in case of positive class and 109 in case of negative one. Boxplot doesn't indicate causation between variables, but proves that the classes have distinct distribution regarding these predictors.

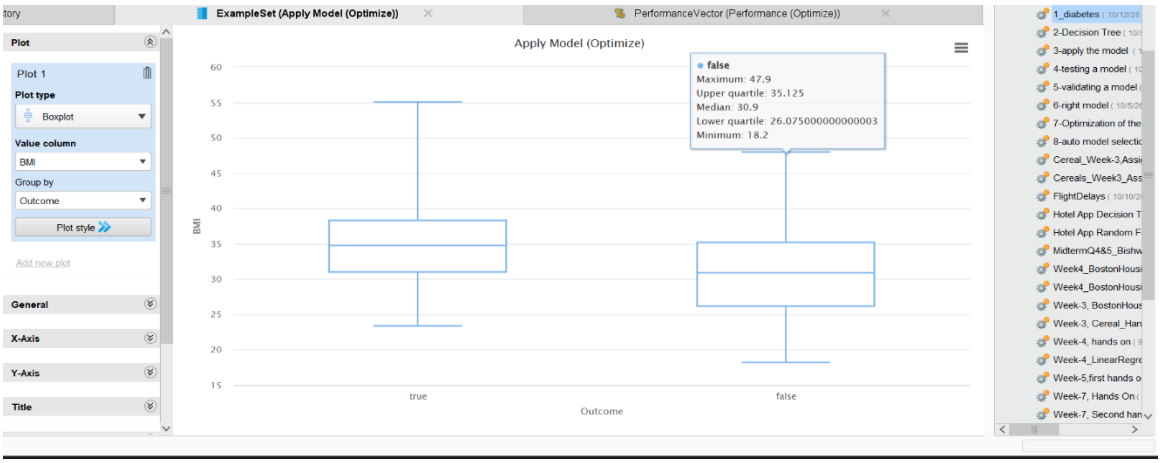


Figure 4. BMI by diabetes Outcome in the held-out ExampleSet.

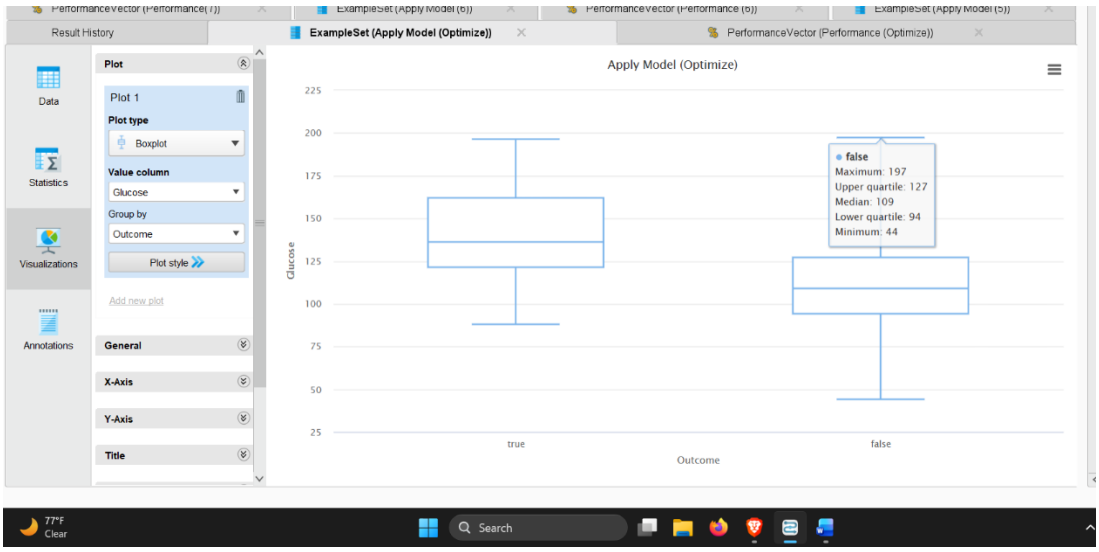


Figure 5. Glucose by diabetes Outcome in the held-out Exampleset.

The Scatter Plot between Glucose and BloodPressure provides us with the two-variable representation. The positive cases of diabetes are generally more frequently found at higher glucose readings. However, bloodpressure alone is not providing any separation. This is consistent with our understanding of classification which should be based on predictors' interplay rather than single predictor's value.

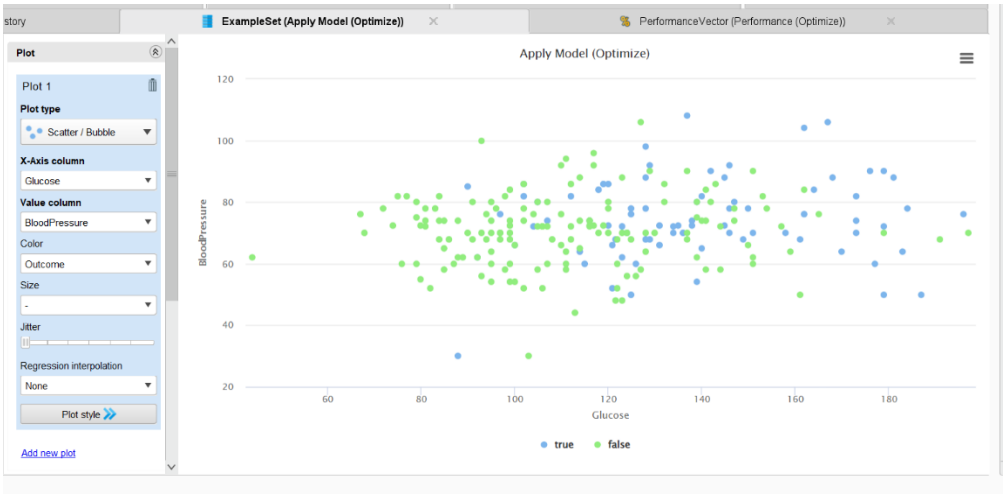


Figure 6. Glucose versus BloodPressure scatter plot colored by Outcome.

3.3 Correlation and Data-Quality Analysis

A Pearson Correlation Matrix for the six attributes was obtained. The strongest positive correlation is observed to be between Pregnancies and Age with r = 0.528. The second strongest correlation is between BloodPressure and Age with r = 0.333. Positive correlation exists between Glucose and Age with r = 0.202. Other correlations are relatively weak. Many of them are zero, such as BloodPressure with DiabetesPedigreeFunction r = -0.003, and BMI with Age r = -0.017. It suggests there are no severe multicollinearity issues for the six attributes shown above.

The correlation graph was obtained from the existing ExampleSet held-out after model execution for transparency purposes. The complete process was maintained without rebuilding. The correlation graph was used for the purpose of descriptive reporting only, not for model training, parameter tuning or model selection.



Figure 7. Correlation heatmap for the six selected attributes.

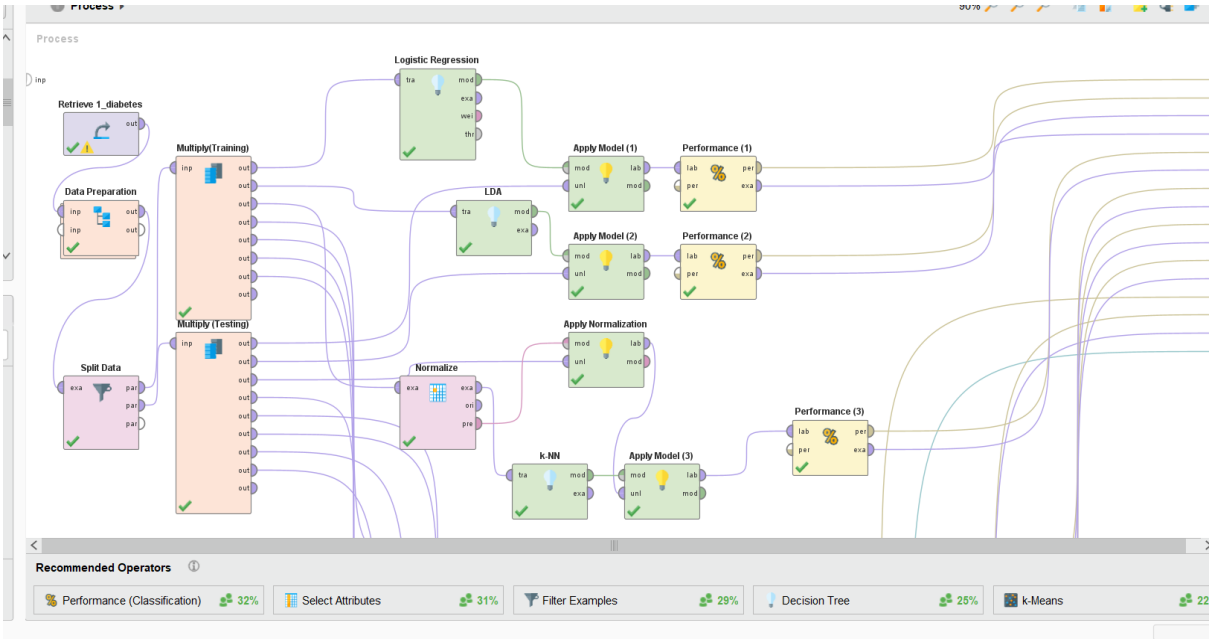
4. Methodology and Modeling

4.1 Model Development

Six classification methodologies were considered using Altair AI Studio. Logistic Regression provided the model. LDA provided a linear decision rule. K-NN provided a non-parametric distance-based methodology. Was tested on normalized data. Naive Bayes provided a baseline model. Random Forest provided bagging by employing an ensemble of decision trees. Gradient Boosted Trees provided boosting by training trees to correct previous errors in sequence. The selected models were chosen in order to compare a linear

model with others making different assumptions concerning decision boundary shapes and relationships between predictors (James et al. 2021).

The trained model was used to test the 230 records hold-out sample. The classifier performance operator provided the following metrics: accuracy, AUC when available, precision, recall, F1 measure, sensitivity, specificity and a confusion matrix. Recall and sensitivity are the same metric here since Outcome = true was the class of interest. LDA did not provide the confidence output needed to calculate an AUC, so LDA's AUC was deemed unavailable.



Picture 8. Final Altair AI Studio process branch showing Logistic regression, Multiply (training), Multiply (Testing).

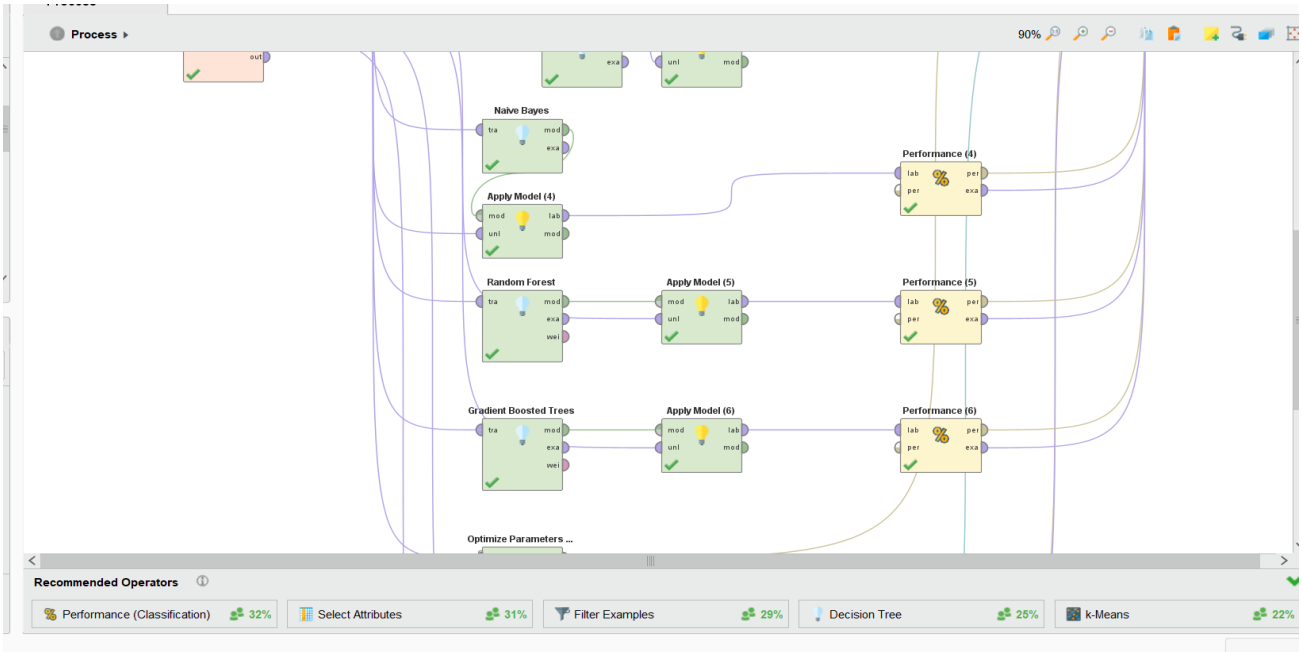


Figure 9. Final Altair AI Studio process branch showing model Naïve Bayes, Random Forest, Gradient Boosted Trees.

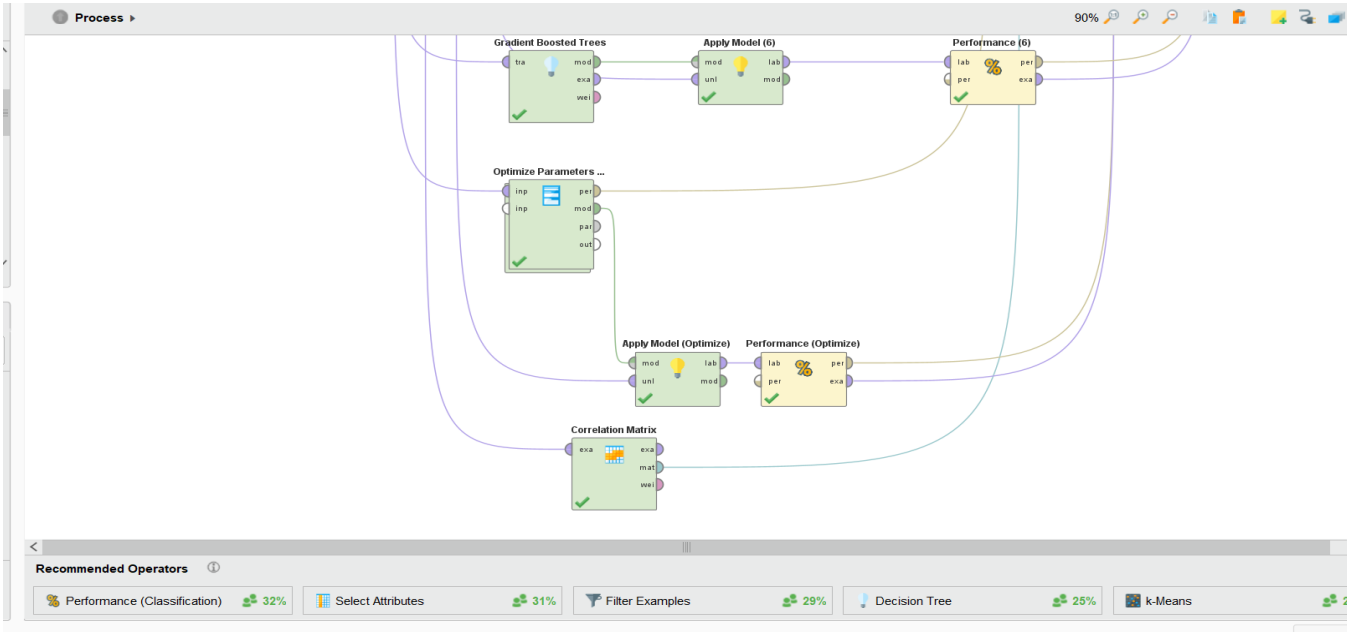
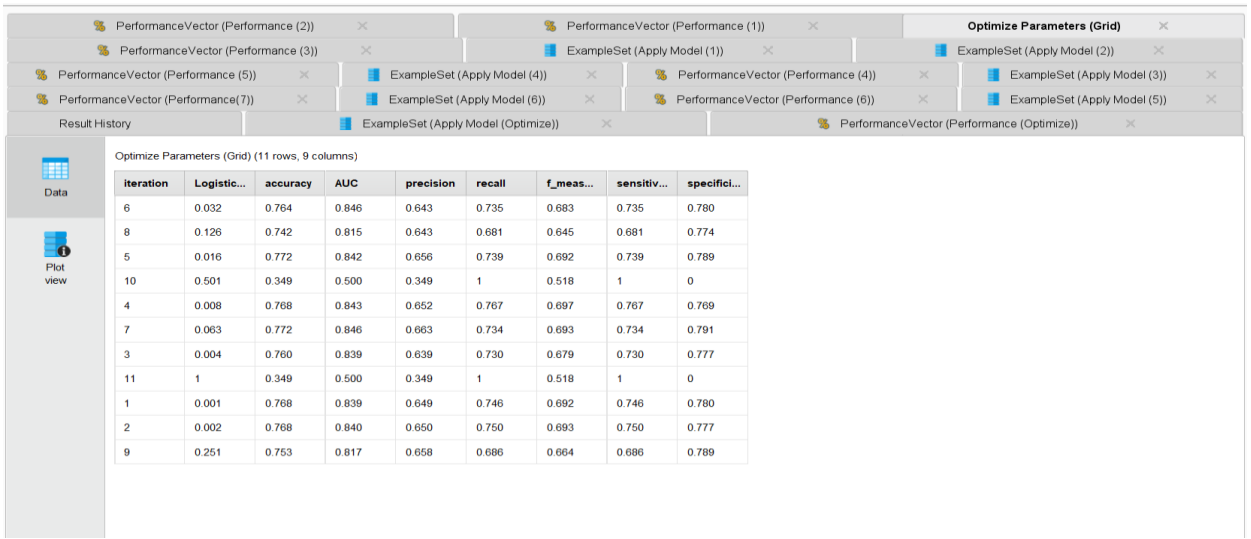


Figure 10. Final Altair AI Studio process branch showing Optimize Parameters, Correlation Matrix.

4.2 Logistic Regression Optimization

The model comparison was followed by logistic regression tuning with the help of Optimize Parameters (Grid). For that, a 10-fold Cross Validation operator was employed. The regularization parameter was tuned within the range of 0.001 to 1, in 11 distinct steps. The accuracy of cross validation for the model with the value of lambda at about 0.016 and 0.063 was 77.2%. According to the cross-validation performance view, the accuracy was 77.16%. The standard deviation was 4.45%.



iteration	Logistic...	accuracy	AUC	precision	recall	f_meas...	sensitiv...	specifici...
6	0.032	0.764	0.846	0.643	0.735	0.683	0.735	0.780
8	0.126	0.742	0.815	0.643	0.681	0.645	0.681	0.774
5	0.016	0.772	0.842	0.656	0.739	0.692	0.739	0.789
10	0.501	0.349	0.500	0.349	1	0.518	1	0
4	0.008	0.768	0.843	0.652	0.767	0.697	0.767	0.769
7	0.063	0.772	0.846	0.663	0.734	0.693	0.734	0.791
3	0.004	0.760	0.839	0.639	0.730	0.679	0.730	0.777
11	1	0.349	0.500	0.349	1	0.518	1	0
1	0.001	0.768	0.839	0.649	0.746	0.692	0.746	0.780
2	0.002	0.768	0.840	0.650	0.750	0.693	0.750	0.777
9	0.251	0.753	0.817	0.658	0.686	0.664	0.686	0.789

Figure 11. Grid-search results for logistic Regression regularization.

This then led to the application of the tuned model on the hold out set. This is due to the fact that the optimal value was chosen through re-sampling of the training data. Therefore, it needed to be applied to the data that was not used in determining the hyperparameters. The output of this model was then compared with the baseline logistic regression model.

5. Performance Evaluation

5.1 Accuracy, Confusion Matrix and ROC/AOC

Holdout Performance Results for All Six Models and the Optimized Logistic Regression Model are listed in Table 2 below. In the Original Models, Logistic Regression Model demonstrated the highest accuracy rate (76.52 percent), the best positive-class recall rate (80.00 percent), the best F1 score (70.33 percent) and achieved AUC value equal to 0.827. This demonstrates balanced performance of the model in terms of positives and negatives.

Model	Accuracy	AUC	Precision	Recall	F1	Specificity
Logistic Regression	76.52%	0.827	62.75%	80.00%	70.33%	74.67%
LDA	73.48%	N/A*	68.63%	43.75%	53.44%	89.33%
k-NN	72.17%	0.772	60.81%	56.25%	58.44%	80.67%
Naive Bayes	72.17%	0.786	60.26%	58.75%	59.49%	79.33%
Random Forest	71.74%	0.804	67.44%	36.25%	47.15%	90.67%
Gradient Boosted Trees	71.74%	0.778	58.06%	67.50%	62.43%	74.00%

Optimized Logistic Regression	74.35%	0.827	62.35%	66.25%	64.24%	78.67%

*Table 2. Holdout/test performance comparison across classification models. *LDA AUC was unavailable in the implemented process because confidence score were not produced.*

Figure 12. Holdout/Test accuracy comparison across models.

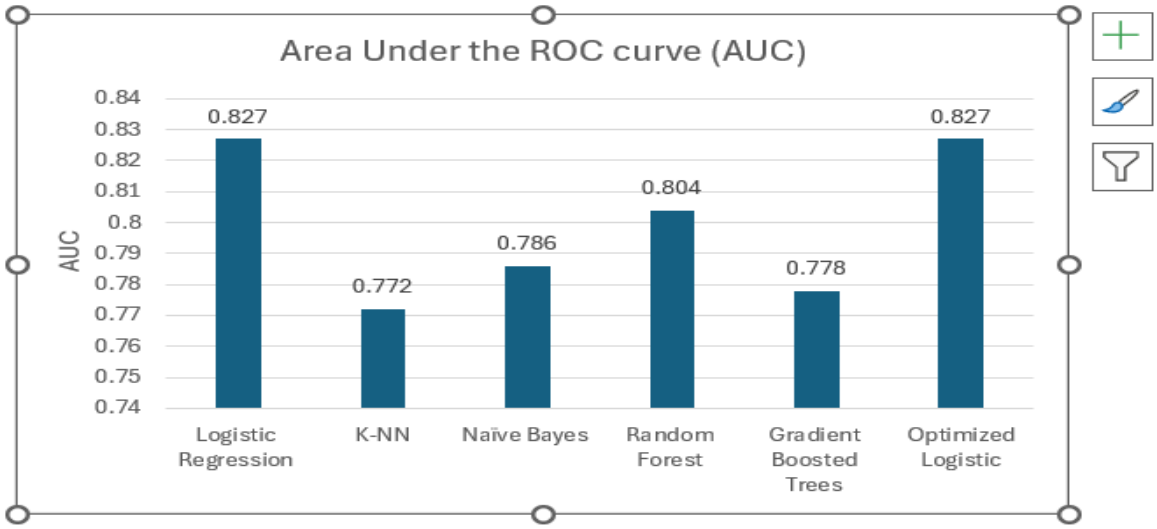


Figure 13. AUC comparison across models with available ROC output.

Confusion Matrix presents the performance of the baseline logistic regression classifier. Confusion Matrix suggests that there are 112 true negative examples out of 150 while there are 38 false negatives out of 150 negative examples. In contrast, 64 positive examples are true out of 80 while 16 positive examples are false out of 80. As a result, a

recall of 80.00% for this class and a specificity of 74.67% for the negative class. ROC AUC of 0.827 indicates good ranking of cases.

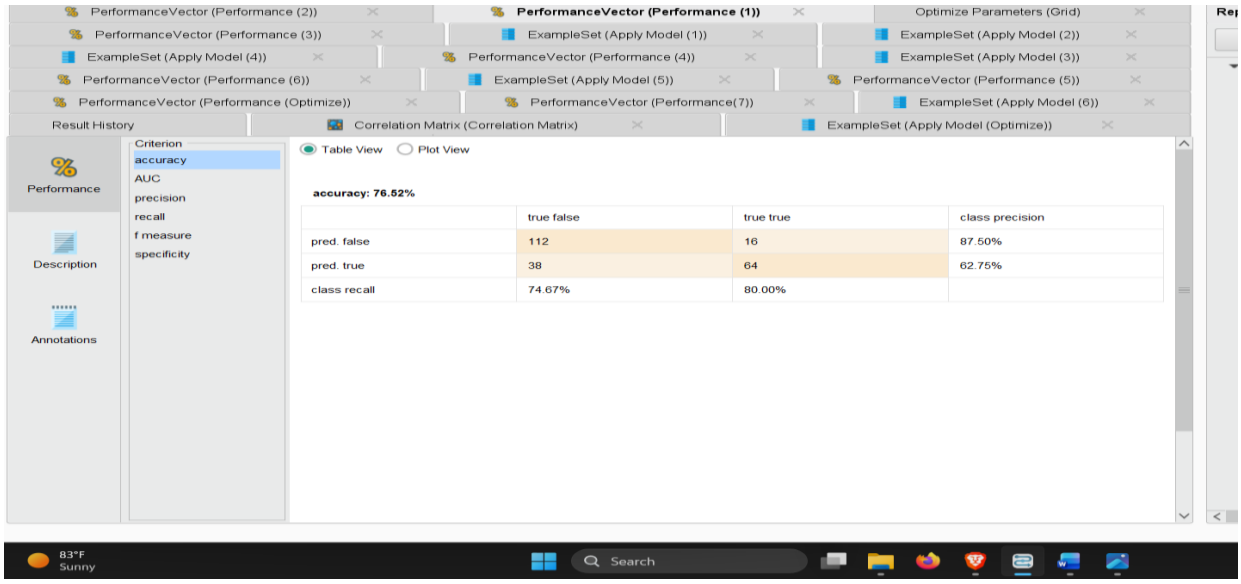


Figure 14. Baseline Logistic Regression confusion matrix and Accuracy.

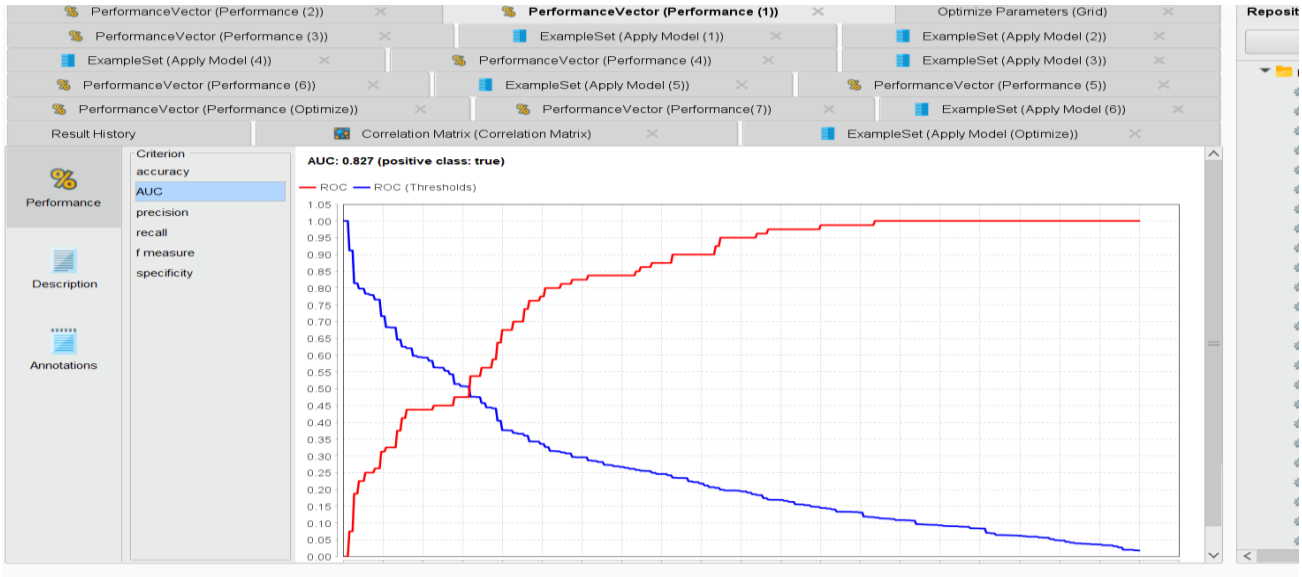


Figure 15. Baseline Logistic Regression ROC and AUC.

5.2 Comparison of Baseline and Optimized Logistic Regression

The accuracy of the optimized model of Logistic Regression was 74.35%. The value of AUC was 0.827. Precision was 62.35% while recall was 66.25%, and the value of F1 score was 64.24% and specificity was 78.67%. As compared to the baseline model, the optimized model improved the performance of the model in terms of specificity by 4% only, decreased recall by 13.75%, and F1 score by 6.09%. AUC remained the same. The optimized model became more conservative and made more false positives, but it could not identify many more positive cases as well.

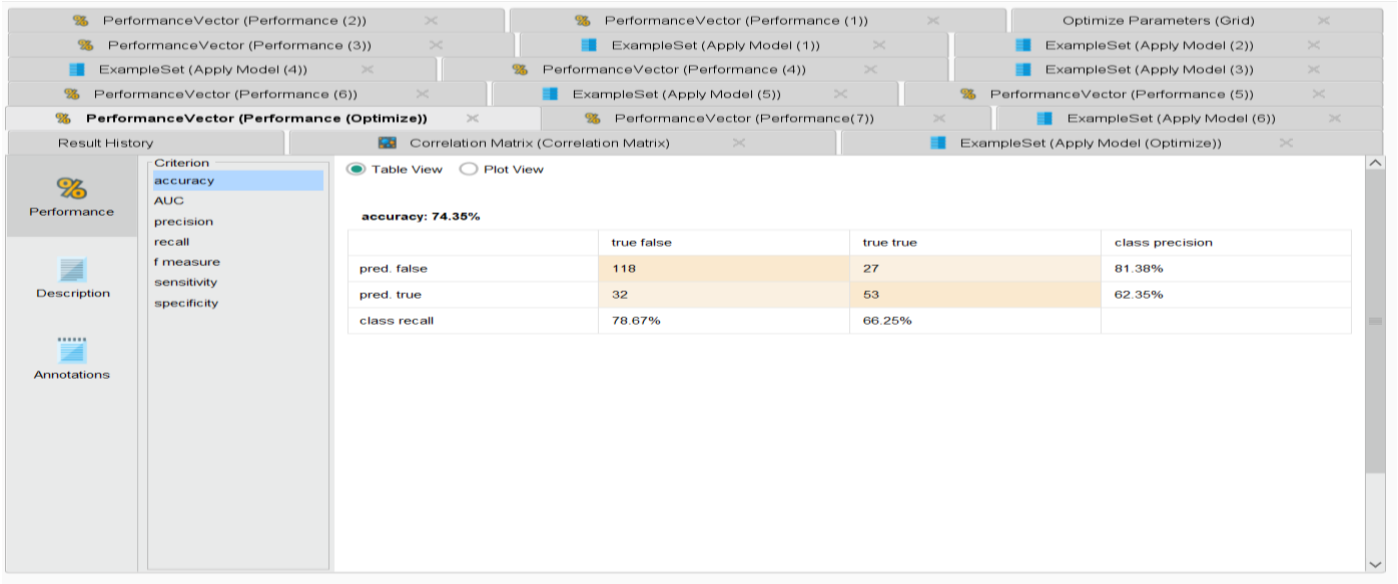


Figure 16. Optimized Logistic Regression confusion matrix and accuracy.

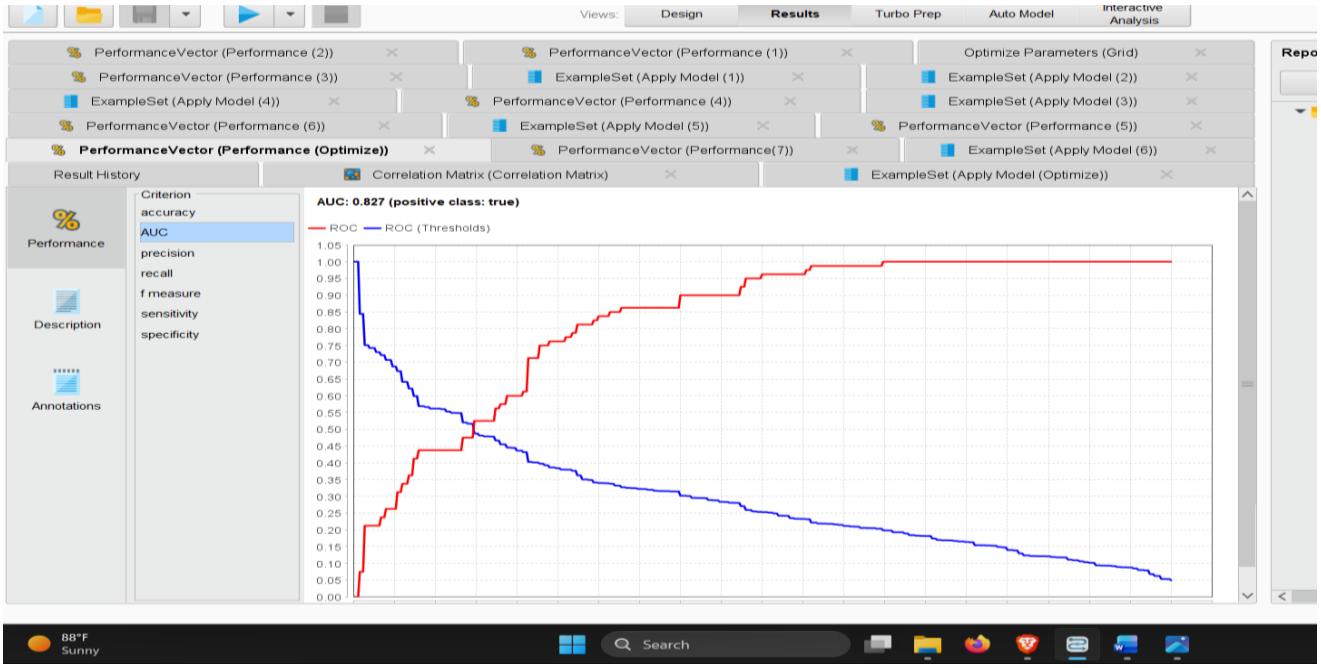


Figure 17. Optimized Logistic Regression ROC curve and AUC.

There is a compromise between the set of models. The Random Forest has the highest specificity at 90.67% and the lowest recall of 36.25%, where there were 51 false negatives out of 80 total positive examples. The Gradient Boosted Trees have the recall of the positive class at 67.50% and a lower accuracy and AUC at 71.74% and 0.778, respectively, as compared to the baseline model of Logistic Regression. LDA has the high specificity and precision but a low recall rate. K-NN and Naive Bayes are better-balanced as compared to Random Forest.

6. Final Model and Insights

Considering the results presented together, the Logistic Regression baseline model is chosen for this project. This model showed the highest test accuracy among six initial

models. It also reached the highest recall. The model showed the highest F1 score. The AUC value was equal to that of another model. In terms of screening analysis, a relatively small amount of false negatives becomes very crucial since a false negative means a positive sample that cannot be detected by the model.

There are several important analytical conclusions that may be drawn based on the analysis performed above. Firstly, it should be stated that the greater complexity of the model did not always lead to successful generalization, since the Logistic Regression model had higher accuracy scores than the tree ensembles when evaluated using the holdout method. Secondly, from the EDA graphs it is possible to conclude that the distribution of glucose and BMI differs between the Outcome groups. From the scatter and correlation analysis it is possible to conclude that no pair of displayed predictors perfectly separates the classes, which means that the prediction task is multivariate.

The project has limitations. The benchmark dataset is small, by machine learning standards. It represents adult women of Pima heritage, which limits direct generalization to other populations. Several predictors also contain a number of invalid zero measurements, especially Insulin and SkinThickness. The chosen imputation method can influence results. Mean imputation was used here because it matched the implemented course workflow. Alternative imputation methods could be compared in work. A larger and more diverse external validation dataset would be needed before any clinical use. Probability calibration, threshold analysis and explicit examination of Logistic Regression coefficients and odds ratios would also improve interpretability.

7. Conclusion

This particular project has provided a complete workflow of machine learning for classifying diabetes risk utilizing Altair AI Studio. The initial phase of the procedure involved taking care of zero values and filling missing measurements with average values. Binary label was formed and 70-30 split was done using stratification. The k-NN branch is. Six algorithms have been trained for the purpose of classification. All six models have been tested using the hold out of 230 records. Logistic Regression model has been optimized using grid search 10 fold cross-validation.

The key result is that the baseline model of logistic regression provided the required balance. The model managed to get 76.52% of accuracy, 0.827 of AUC, 80.00% of recall and 70.33% of F1. The optimized model of logistic regression provided higher specificity but reduced the recall and F1 on the holdout sample. Random Forest also confirmed that high specificity does not necessarily equal detection of positive cases. The key lesson from these results is that it is important to consider metrics and confusion matrix.

On the whole, the project has been successful in building, comparing and optimizing models of diabetes risks. The recommended model for the given workflow is thus the Logistic Regression model because it was the best performing one. This model should be viewed as a prediction tool and not a clinical diagnosis tool.

References

mragpavank. (n.d.). Pima Indians diabetes database [Data set]. Kaggle.

https://www.kaggle.com/datasets/mragpavank/diabetes

Appendix A. Confusion Matrices

Table 3 shows the confusion matrix counts, for the 230-record holdout set. The testing set had 150 negative cases and 80 actual positive cases. These counts directly support the recall and specificity values that were reported.

Model	TN	FP	FN	TP
Logistic Regression	112	38	16	64
LDA	134	16	45	35
k-NN	121	29	35	45
Naive Bayes	119	31	33	47
Random Forest	136	14	51	29
Gradient Boosted Trees	111	39	26	54
Optimized Logistic Regression	118	32	27	53

Table 3. Confusion-matrix counts for the 230-record holdout set.

TN = true negative; FP = false positive; FN = false negative; TP = true positive.

Appendix B. Final Process and Submission Notes

Submission through Canvas must have Word report with ML Project Plan template, final process file (.rmp) from Altair AI Studio/RapidMiner together with the sample data set/data source utilized. Also, the assignment requires that there is a PowerPoint slide for presentation. Prior to the submission of the .rmp process file, it should first be saved having ensured that all model operators, optimization operators, performance operators, and correlation operators do not have any connections.

Recommeded file naming follows the course convention:

CSBUX4_BISHNUBishwokarma.doc

CSBUX4_BISHNUBishwokarma.ppt